Assignment: Correlation

Fitness Data from Men in a Physical Fitness Course at N.C. State University

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Introduction:

Based on this data assignment, that contains fitness dataset have physiological measurements from men taking in physical fitness course at N.C. State University. The data variable contains age, weight, oxygen, run time, rest pulse, run pulse, and maximum of pulse. Based in this data measure, it gives some insights on how the individual's cardiovascular and physical fitness levels to ensure how they are doing. It can be study relationships between body characteristics and fitness performance. Therefore, it is important to know the difference based on the factors that influence on heart rate, oxygen, consumption, and overall fitness capacity.

Data Audit / Data Overview:

In these datasets, the variables that contains are age, weight, oxygen (oxy), run time (runtime), rest pulse (rstpulse), run pulse (runpulse), and maximum of pulse (maxpulse).

Variables	Describing the variables
Age (age)	Participant's age
Weight (weight)	Participant's Weight
Oxygen (oxy)	Oxygen consumption
Run Time (runtime)	Duration of run time
Rest Pulse (rstpulse)	Rate of resting pulse
Run Pulse (runpulse)	Rate of pulse while running
Maximum of Pulse (maxpulse)	Maximum pulse rate

Figure 1: Tables of Describing the Variables

Alphabetic List of Variables and Attributes			
#	Variable	Type	Len
1	age	Num	8
8	case	Num	8
7	maxpulse	Num	8
3	oxy	Num	8
5	rstpulse	Num	8
6	runpulse	Num	8
4	runtime	Num	8
2	weight	Num	8

Figure 2: Output of SAS for List of Variables and Attributes

Based on the data overview where it will show in the first 10 observation as follows:

Overview of Fitness Dataset in the first 10 Observations

Obs	age	weight	oxy	runtime	rstpulse	runpulse	maxpulse	case
1	44	89.47	44.609	11.37	62	178	182	1
2	40	75.07	45.313	10.07	62	185	185	2
3	44	85.84	54.297	8.65	45	156	168	3
4	42	68.15	59.571	8.17	40	166	172	4
5	38	89.02	49.874	9.22	55	178	180	5
6	47	77.45	44.811	11.63	58	176	176	6
7	40	75.98	45.681	11.95	70	176	180	7
8	43	81.19	49.091	10.85	64	162	170	8
9	44	81.42	39.442	13.08	63	174	176	9
10	38	81.87	60.055	8.63	48	170	186	10

Figure 3: Output of Fitness Dataset in the first 10 observation

<u>Hypothesis Test for a Correlation:</u>

Each pair of variables tested, the hypothesis would be:

$$H_0: \rho = 0: There\ is\ no\ linear\ relationship\ between\ two\ variables.$$

$$H_a: \rho \neq 0: There\ is\ a\ linear\ relationship\ between\ two\ variables.$$

The decision based on the p-value with the significant value of 0.05 ($\alpha = 0.05$):

- If $\rho < 0.05$, reject null hypothesis and conclude that there are significant correlations.
- If $\rho > 0.05$, failed to reject null hypothesis and conclude that there are no significant correlations.

Correlation:

Correlations of Fitness Data

The CORR Procedure

7 Variables:	age weight oxy runtime rstpulse runpulse maxpulse

Simple Statistics						
Variable	N	Mean	Std Dev	Sum	Minimum	Maximum
age	31	47.67742	5.21144	1478	38.00000	57.00000
weight	31	77.44452	8.32857	2401	59.08000	91.63000
oxy	31	47.37581	5.32723	1469	37.38800	60.05500
runtime	31	10.58613	1.38741	328.17000	8.17000	14.03000
rstpulse	31	53.45161	7.61944	1657	40.00000	70.00000
runpulse	31	169.64516	10.25199	5259	146.00000	186.00000
maxpulse	31	173.77419	9.16410	5387	155.00000	192.00000

Figure 4: SAS Output of Table for Correlations of Fitness Data

Spearman Correlation Coefficients, N = 31 Prob > |r| under H0: Rho=0							
	age	weight	oxy	runtime	rstpulse	runpulse	maxpulse
age	1.00000	-0.16152 0.3853	-0.18351 0.3231	0.15883 0.3934	-0.11766 0.5285	-0.29810 0.1033	-0.38682 0.0316
weight	-0.16152 0.3853	1.00000	-0.09318 0.6181	0.07483 0.6891	-0.02958 0.8745	0.07541 0.6868	0.14260 0.4441
oxy	-0.18351 0.3231	-0.09318 0.6181	1.00000	-0.80806 <.0001	-0.38028 0.0348	-0.43748 0.0138	-0.32239 0.0769
runtime	0.15883 0.3934	0.07483 0.6891	-0.80806 <.0001	1.00000	0.48618 0.0056	0.28390 0.1217	0.20580 0.2667
rstpulse	-0.11766 0.5285	-0.02958 0.8745	-0.38028 0.0348	0.48618 0.0056	1.00000	0.36765 0.0419	0.32634 0.0732
runpulse	-0.29810 0.1033	0.07541 0.6868	-0.43748 0.0138	0.28390 0.1217	0.36765 0.0419	1.00000	0.93152 <.0001
maxpulse	-0.38682 0.0316	0.14260 0.4441	-0.32239 0.0769	0.20580 0.2667	0.32634 0.0732	0.93152 <.0001	1.00000

Figure 5: SAS Output of Pearson Correlations of Fitness Data

Pearson Correlation Coefficients, N = 31 Prob > |r| under H0: Rho=0							
	age	weight	oxy	runtime	rstpulse	runpulse	maxpulse
age	1.00000	-0.23354 0.2061	-0.30459 0.0957	0.18875 0.3092	-0.16410 0.3777	-0.33787 0.0630	-0.43292 0.0150
weight	-0.23354 0.2061	1.00000	-0.16275 0.3817	0.14351 0.4412	0.04397 0.8143	0.18152 0.3284	0.24938 0.1761
oxy	-0.30459 0.0957	-0.16275 0.3817	1.00000	-0.86219 <.0001	-0.39936 0.0260	-0.39797 0.0266	-0.23674 0.1997
runtime	0.18875 0.3092	0.14351 0.4412	-0.86219 <.0001	1.00000	0.45038 0.0110	0.31365 0.0858	0.22610 0.2213
rstpulse	-0.16410 0.3777	0.04397 0.8143	-0.39936 0.0260	0.45038 0.0110	1.00000	0.35246 0.0518	0.30512 0.0951
runpulse	-0.33787 0.0630	0.18152 0.3284	-0.39797 0.0266	0.31365 0.0858	0.35246 0.0518	1.00000	0.92975 <.0001
maxpulse	-0.43292 0.0150	0.24938 0.1761	-0.23674 0.1997	0.22610 0.2213	0.30512 0.0951	0.92975 <.0001	1.00000

Figure 6: SAS Output of Spearman Correlations of Fitness Data

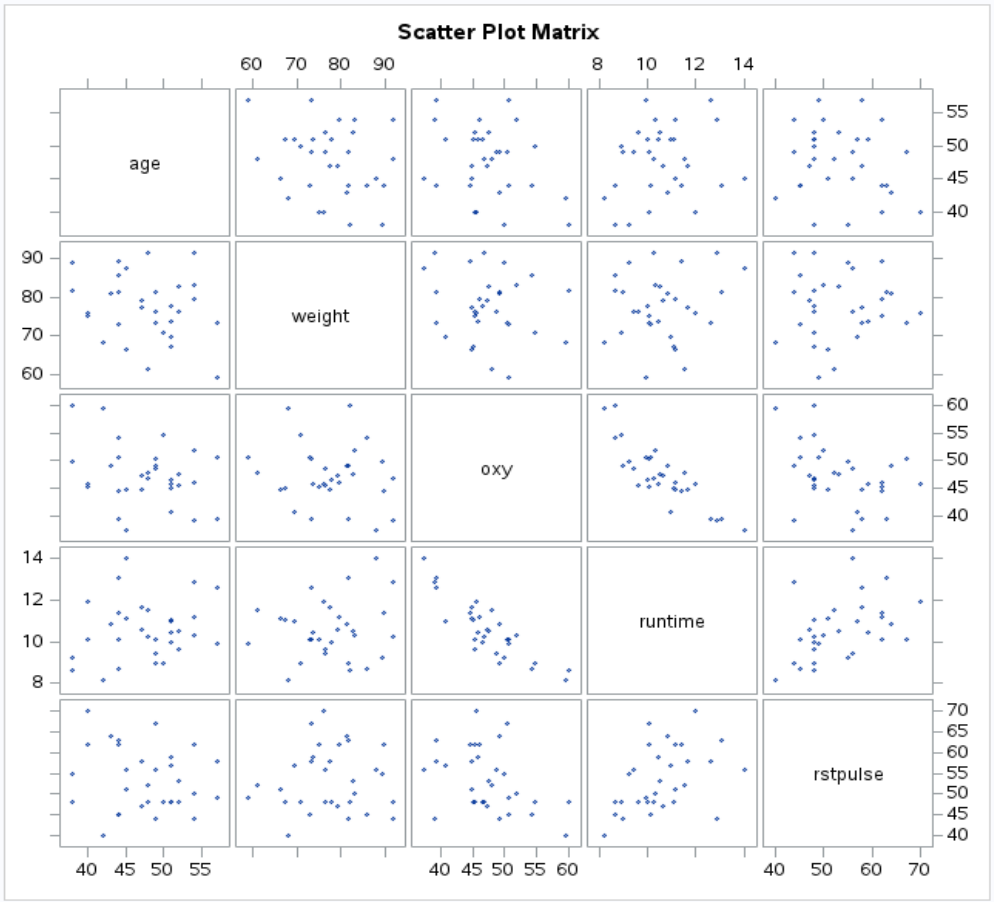

Figure 7: SAS Output of Scatter Plot Matric for Fitness Data

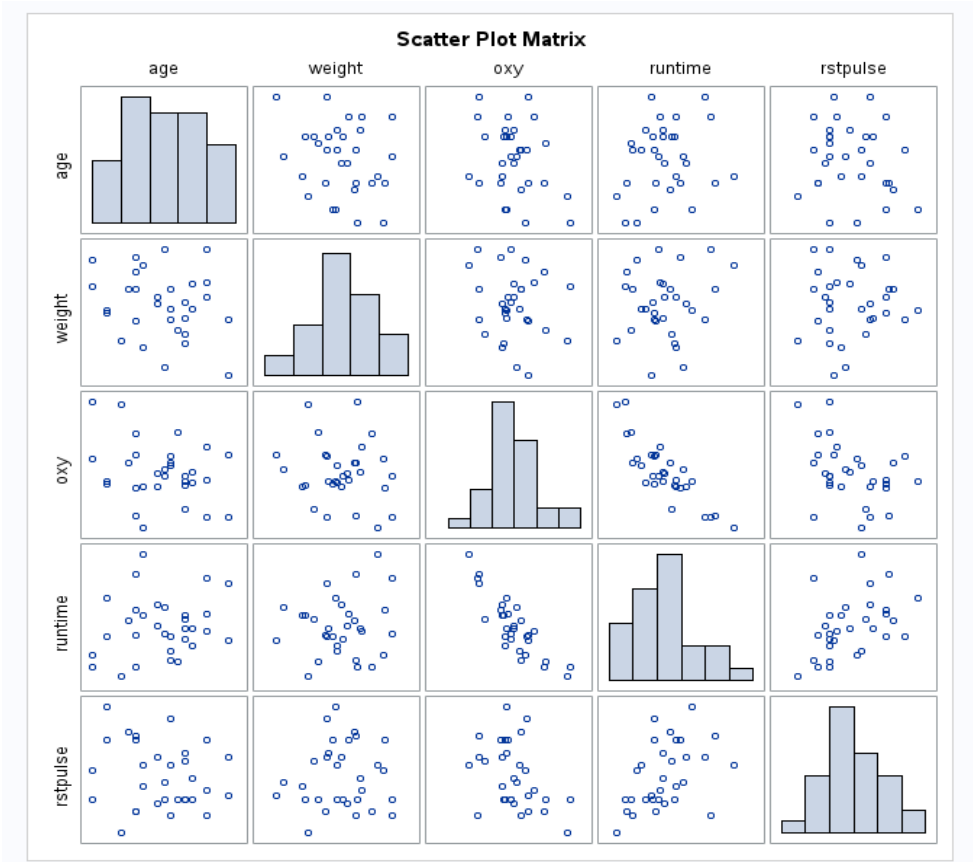

Figure 8: SAS Output of Scatter Plot Matrix with Histogram for Fitness Data

<u>Key Findings:</u>

The key findings for correlation on this fitness data based on the **Figure 5: SAS Output of Pearson Correlation of Fitness Data** and **Figure 6: SAS Output of Spearman Correlation of Fitness Data** tells that one of correlation have strong positive between runpulse and maxpulse with the correlation coefficient of 0.92975 in Pearson Correlation and 0.93152 in Spearman Correlation.

Other thing, there is strong negative correlation between runtime and oxy with the correction coefficient of -0.86219 in Pearson Correlation and -0.80806 in Spearman Correlation based on the Figure 5 and Figure 6.

And the rest of induvial variables, it is showing weak correlations coefficient as it is showing $0 < r < 0.5$ based on the results of correlation coefficient.

As it can see in figure 6 and 7, there are the matrix of scatterplot which includes histogram options where you find runpulse and maxpulse showing the plots have upwards trends which shows strong positive correlations while runtime vs oxy with plots have downwards trends that

shows negative positive correlations. Other variables, it is showing all the plots are everywhere which indicates weak correlations.

Other than, any p-value which is less than 0.001 (<0.001) for correlations shows it is statistically significant.

<u>Two Individual Scatter Plots (Strong Positive Correlation and Strong Negative Correlation):</u>

Strong Positive Correlation

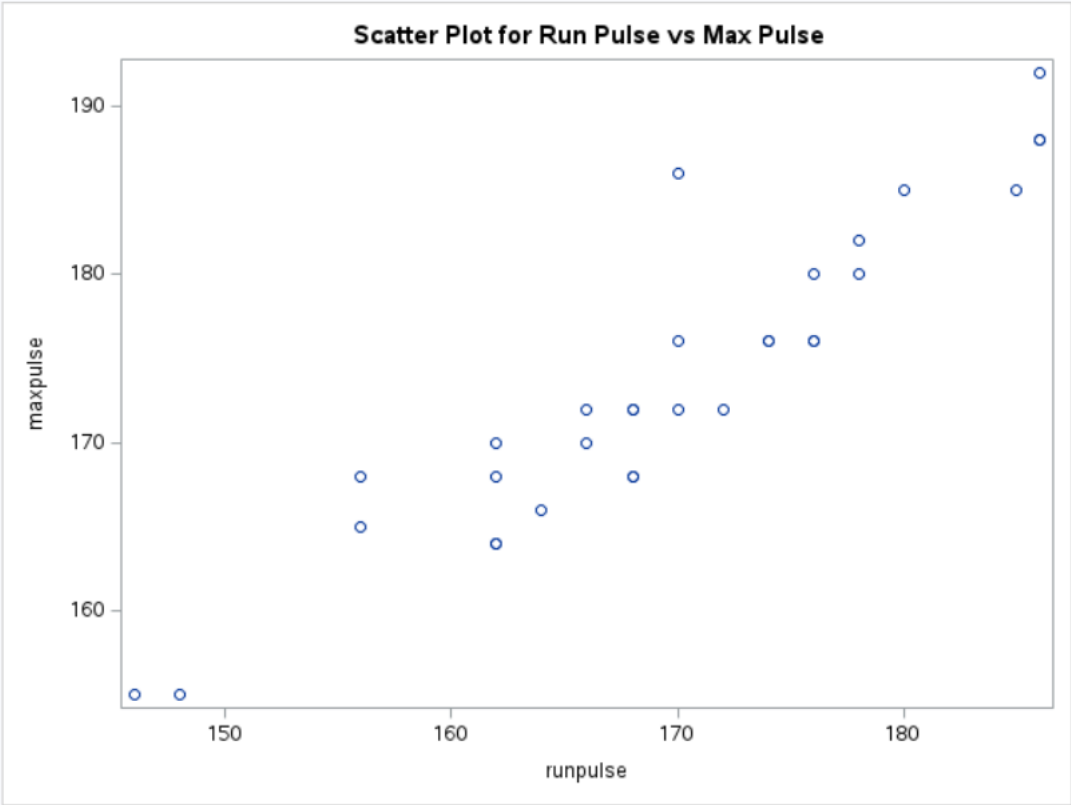

Figure 9: Scatter Plot for Run Pulse vs Max Pulse

In **Figure 9: Scatter Plot for Run Pulse vs Max Pulse** is showing positive correlation where you can see the several dots are upwards trends. It suggests that pulse rate increase, then the maximum pulse rate could rise up. The reason because of working hard which able to meets body's oxygen that need to reach maximum heart rate which can cause run pulse and max pulse increases. Therefore, it is showing cardiovascular exertion.

Strong Negative Correlation

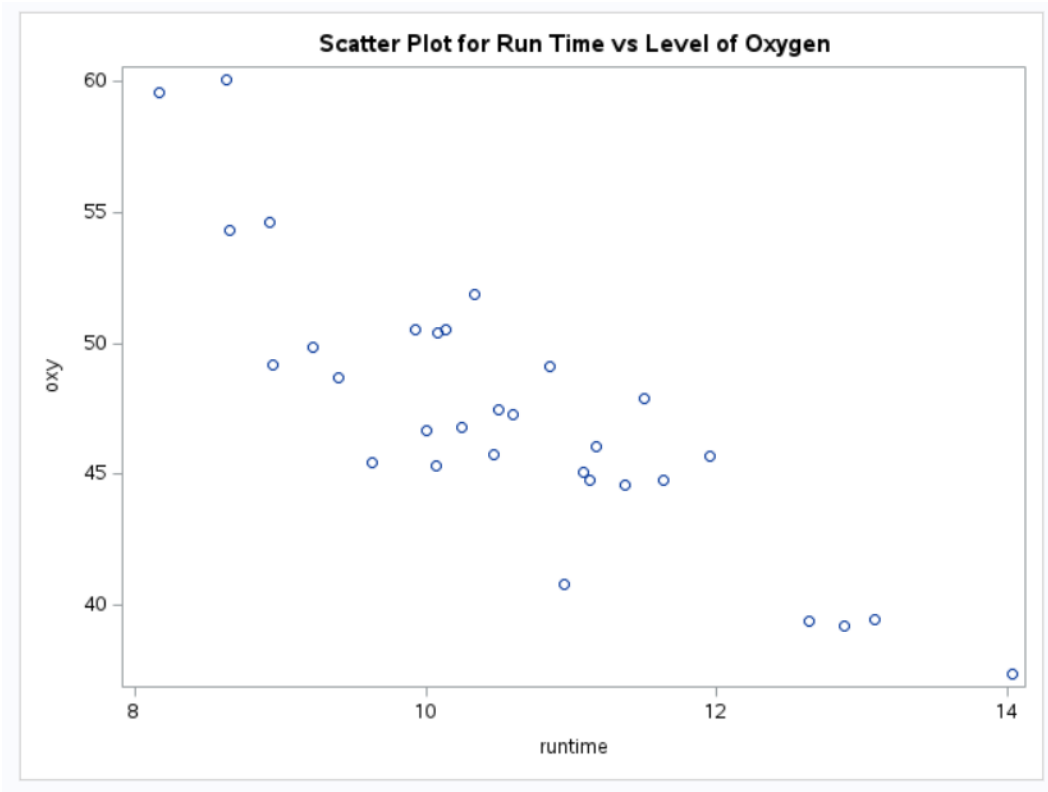

Figure 10: Scatter Plot for Run vs Level of Oxygen

In Figure 10: Scatter Plot for Run vs Level of Oxygen, it is showing strong negative correlation where the plot are showing downwards trends which implies that participants with shorter run time may have high level of oxygen consumptions while the participants with longer run time could lower the oxygen consumption levels. It can be due to muscle requires working hard, produce more oxygen, breathing rate and depth increase more oxygen so if participants run during the time, it may reduce their breath. It is showing better cardiovascular fitness.

<u>Limitations of the Correlation Coefficient:</u>

The limitations of the correlation coefficient show the relationships between variables which includes age, weight, levels of oxygen, running pulse, resting pulse, and maximum pulse, however, it may not appear any non-linear patterns may exist. Based on the strong positive correlation like Run Pulse vs Max Pulse and strong negative correlation as Run time vs Oxygens does not mean completely strong correlation as there might be other factors that would affect the results. So therefore, correlation does not imply the action. However, there might be an additional variables like third variables which may relationship stronger or weaker correlation. For the Outliers, the participants sometimes can be showing as high or low pulse rate which can strengthen the correlation. For the sample size, it can affect the p-value especially with small dataset and moderate correlations shows statistically significant, but it can affect if the third

variables added. Therefore, it is showing association, not cause-and-effort and it should interpret carefully to get accurate results when analyzing physical fitness performance.

Conclusion:

In conclusion, based on the findings and two induvial scatter plots, the fitness data shows run pulse vs max pulse have strong positive correlation which means participant's pulse while running could increase while the maximum pulse also rise as the same time. For run time vs level of oxygen showing strong negative correlation, indicates that participants running in less time which could have higher levels of oxygen while the participants running with more times could lower the levels of oxygen. Other than, the rest of variable are showing weak correlation coefficient where the plots are showing everywhere. However, this dataset could limit the findings which might showing inaccuracy results. For the next steps, matrix correlation provides comprehensive view that indicates where it shows every variable and where it the plots going, highlighting with careful interpretation of correlation coefficients in this fitness data.

Works Cited

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